discourse to be discarded (Desideratum 2). For Desideratum 2, we furthermore considered experiments using operation descriptions with greater context dependence. Specifically, we tested scenarios with an additional *Move contents of Box N to Box M* operation that does not explicitly mention the object names, and scenarios where object descriptions in the operations can only be fully disambiguated by knowing the current state of a box. For Desideratum 3, we considered experiments where the phrasing of the states and operations differ entirely between demonstration/finetuning and evaluation (see Table 2). Finally, for all experiments, we computed a "signature" of every initial state that indicates the number of objects contained in each box.[4] Then, we ensured that there were no two examples with identical initial descriptions modulo the object names where one appeared in the training split and the other one in the evaluation split. This prevents this task from being solvable via slot-filling (Desideratum 4).[5]

3.3 Task

We define the entity tracking task as follows. Given a natural language description of the initial state of the world followed by 0–12 state-changing operations, the content of each box at the end of the description must be correctly identified. To evaluate this, we created a test example for each box after each operation. This corresponds to $n \times (\text{NumOps} + 1)$ examples per scenario (91 exs. in our datasets). Each example is formulated in the style of a cloze test. That is, the input describes the initial state followed by a sequence of operations, ending in *Box N contains* __. The expected output is the correct set of objects in Box N based on the prefix description. See Appendix B for an example.

4 Experiment 1: In-context Demonstration

In the first set of experiments, we evaluated pretrained LMs using a small number of in-context demonstrations of the entity tracking task. This provides a way to probe the model without requiring substantial supervision for the task, as well as guiding the model to output the final state in a consistent format that can be automatically assessed.[6]

4.1 Models

We used models that are known to support task adaptation via in-context demonstrations: GPT-3 175B (`davinci`: Brown et al. 2020), GPT-3.5 (`text-davinci-003`[7]), and Flan-T5 (base and XL: Chung et al. 2022). The little information that OpenAI revealed about their models[8] suggests that `davinci` is an autoregressive language model primarily trained on text corpora. `text-davinci-003` was trained on the language modeling objective on a mix of text and code, and additionally tuned with human feedback using reinforcement learning. Flan-T5 is based on T5, a sequence-to-sequence model trained on a denoising objective, that has been further instruction-finetuned on a battery of tasks. This has been shown to promote better responses to instructions, both with and without demonstrations (Chung et al., 2022). We evaluated the GPT models through the OpenAI API and the Flan-T5 using the HuggingFace library (Wolf et al., 2020). See Table 1 for a summary of the models, and Appendix C for implementation details.

We compared these models against a baseline that randomly outputs 0 to $m = 3$ objects from the set of objects that appeared in the same clauses as the box in question. Note that this baseline is much stronger than a fully random baseline that selects outputs from all mentioned objects.

4.2 Prompting and Demonstrations

Our prompts consist of: (a) a general instruction for the task, (b) two examples of the task to demonstrate the expected format, (c) an initial state description followed by a series of operations, and (d) an incomplete sentence *Box N contains* ___ that the model should complete (see Appendix E for full prompts). We used demonstrations that output the state of all boxes at once. However, in early experiments, Flan-T5 frequently only output the state of the first box even when the in-context demonstrations contained final descriptions of all box states.

[4] For example, the signature of an initial state in which the first box contains two objects and the rest contains 1 object each would be 2111111.

[5] Additionally, compared to the Alchemy setup, our setup also has a benefit of requiring fewer additional reasoning abilities. The beaker domain in Alchemy requires the model to count and perform simple arithmetic (e.g., adding one unit of liquid followed by adding two units of liquid results in three units of liquid). Moreover, some of the operations in Alchemy require knowledge about how colors are combined (e.g., mixing red and green liquids results in a brown liquid). The boxes domain removes these requirements.

[6] For GPT-3.5, we were able to instruct the model to output predictions in a consistent format without any in-context demonstrations, thus making it a true zero-shot experiment. We discuss the results of this experiment in Appendix D.

[7] https://beta.openai.com/docs/models/gpt-3

[8] https://beta.openai.com/docs/model-index-for-researchers

Model	Size	Code?	Additional Training
GPT-3 `davinci`	175B	✗	-
GPT-3 `davinci-instruct-beta`	175B	✗	Human demonstrations (finetuning)
GPT-3 `text-davinci-001`	175B	✗	Human demonstrations + highly rated model outputs (finetuning)
GPT-3.5 `code-davinci-002`	?	✓	-
GPT-3.5 `text-davinci-002`	?	✓	Human demonstrations + highly rated model outputs (finetuning)
GPT-3.5 `text-davinci-003`	?	✓	RL on human feedback
Flan-T5 base	250M	✗	1.6K tasks + instructions (finetuning)
Flan-T5 XL	3B	✗	1.6K tasks + instructions (finetuning)

Table 1: Summary of the models used for the in-context demonstration experiments.

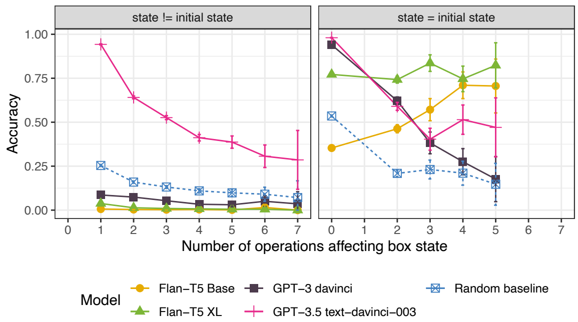

Figure 2: Accuracy on state prediction after n operations that affect a specific box. Left: predictions for boxes whose content differs from the initial state, Right: predictions for boxes whose content is the same as in the initial state. Error bars show 95% CIs.

Therefore, for Flan-T5, we adjusted the prompts to output each box individually.[9]

Demonstration/Test Mismatch for Form-meaning Disentanglement (AltForms) As discussed in Sections 3.1–3.2, we additionally experimented with a setup where the demonstration and test examples were mismatched in the form of the descriptions of the states and operations. Under this setup, the models were evaluated with set of object names and phrasings of the state and operation descriptions that were different from the demonstration examples (see Table 2). Except for the determiner *the* and the preposition *into*, the two sets share no words (although subwords may be shared depending on tokenization).

Scenarios with Greater Context Dependence (MoveContents, AmbiRef) As also discussed in Sections 3.1–3.2, we experimented with two additional scenarios with greater context dependence. The first scenario (MoveContents) introduces an additional operation *Move contents of Box N to Box M* that moves all objects in Box N to Box M, that does not provide an explicit enumeration of objects being moved. This requires the model to rely on the preceding description to identify the set of objects being moved, further removing room for heuristics that allow the prediction of the final state without composing the initial description and the operations in temporal order. The second scenario (AmbiRef) adds adjectival modification to object names (e.g., *the big brain* and *the small brain*), where the adjective can be omitted in some of the operations, depending on the state of the box being described. Specifically, the modifier is dropped if there is only one object of a specific type in a box (e.g., *Move the brain from Box 1 to Box 2* if there is only one brain in Box 1). When predicting the contents of a box, the model is instructed to output the object type with the correct adjective to fully disambiguate the referring expression for the object. This again requires composition of the initial description and the operations in temporal order to correctly interpret the otherwise ambiguous object mentions. (See Appendix B for example scenarios and operations.)

4.3 Evaluation

We estimated the entity tracking capacity of the models by computing the accuracy of predicting the contents of each box after each operation. Given that we rely on arbitrary-length cloze completion to predict the contents, we had to score unconstrained generations. While we only considered instances as correct where the output mentions all objects (and no additional objects) in a given box, our evaluation allowed for minor deviations from the exact form of the response. Namely, we allowed the objects to appear in any order, the object names may be separated by commas or *and*, and both complete noun phrases with a determiner (e.g., *the furby*) and bare nouns (*e.g., furby*) are considered correct.

The task becomes intrinsically harder as more operations are applied to a specific box, since the initial state description needs to be combined sequentially with all subsequent operations. Further, similarly to our observation in Section 2, every operation changes the state of at most two boxes.

[9] To verify that this difference in the task format does not underestimate the accuracy of the GPT models, we also conducted an experiment in which we prompted GPT to only output the state of one box at a time. We found that, contrary to the Flan-T5 models, the accuracy of GPT was *lower* when we prompted it to output individual boxes than prompting it to output all boxes, so we can rule out that this difference in the task format is underestimating GPT's performance.

Operation	Base	AltForms
`Move`	*Move the car from Box 1 to Box 3.*	*Pick up the furby in Container A and place it into Container C.*
`Remove`	*Remove the car from Box 1.*	*Take the furby out of Container A.*
`Put`	*Put the car into Box 1.*	*Place the furby inside Container A.*

Table 2: Different phrasings of the state-changing operations under the AltForms evaluation setup.

This implies that the number of datapoints corresponding to fewer operations is much greater than the number of datapoints for more operations. For these reasons, we report accuracy as a function of the number of operations affecting a particular box rather than reporting aggregate accuracy, and show all results with 95% confidence intervals.

4.4 Results

Figure 2 shows the prediction accuracy for different number of operations that affected a box (e.g., 3 indicates that three operations changed the content of the box after the initial state). The left panel shows the instances where the probed state differed from the initial state; the right panel shows the instances where the probed state was the same as the initial state. As the left panel shows, only GPT-3.5 `text-davinci-003` consistently outperformed the (strong) random baseline. While, not surprisingly, the accuracy of this model also decreases as the number of operations increases, it still correctly predicted all contents of a box after 7 operations in more than 25% of the cases. The Flan-T5 models, on the other hand, seemed to ignore the operations and primarily predicted the initial state description, as indicated by the consistently high accuracy when the final state matches the initial state (right panel), as well as the consistently low accuracy when the final state deviates from the initial state (left panel). GPT-3 `davinci` also primarily repeated the initial state, but as indicated by the steep decrease in the right panel, it was distracted by intervening operations even when repeating the initial state.

Form-meaning Disentanglement We additionally evaluated `text-davinci-003`, the only model that exhibited a non-trivial entity tracking capacity in the first set of results, under the AltForms setup as described in Section 4.2 where the demonstration examples have low lexical overlap with the test examples. Figure 3 shows the prediction accuracy of `text-davinci-003` on a representative subsample[10] of our data. The blue line represents the performance when the descriptions in the demonstration and the test examples have low lexical overlap. As the comparison to the original results (red line) shows, the disjoint demonstrations did lead to a small drop in performance when there were more than two operations affecting a box. Nevertheless, `text-davinci-003` was able to predict the correct state of entities in many cases, further adding support for its non-trivial entity tracking capacity.

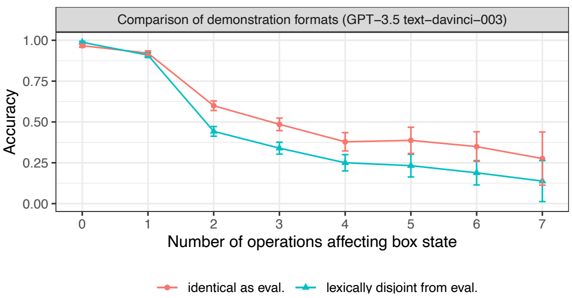

Figure 3: Entity tracking accuracy of `text-davinci-003` with low lexical overlap between demonstration and test examples (AltForms).

Greater Context Dependence Finally, we evaluated `text-davinci-003` on two additional datasets described in Section 4.2: the AmbiRef dataset where adjectival modification can be omitted depending on the current context, and the MoveContents dataset where a new operation *Move contents of Box N to M* that requires the contents of Box N to be contextually identified. Figure 4 shows the performance of `text-davinci-003` on these two datasets compared to our random baseline. As

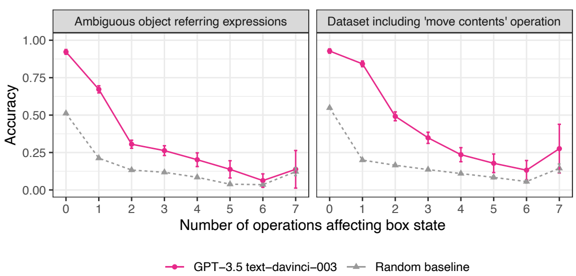

Figure 4: Entity tracking accuracy of `text-davinci-003` for the AmbiRef (left) and MoveContents (right) datasets.

[10]For each number of operations n affecting a box, we sampled 100 states with at least one example with n operations.